



**សាកលវិទ្យាល័យអន្តរជាតិ
PHNOM PENH INTERNATIONAL UNIVERSITY**

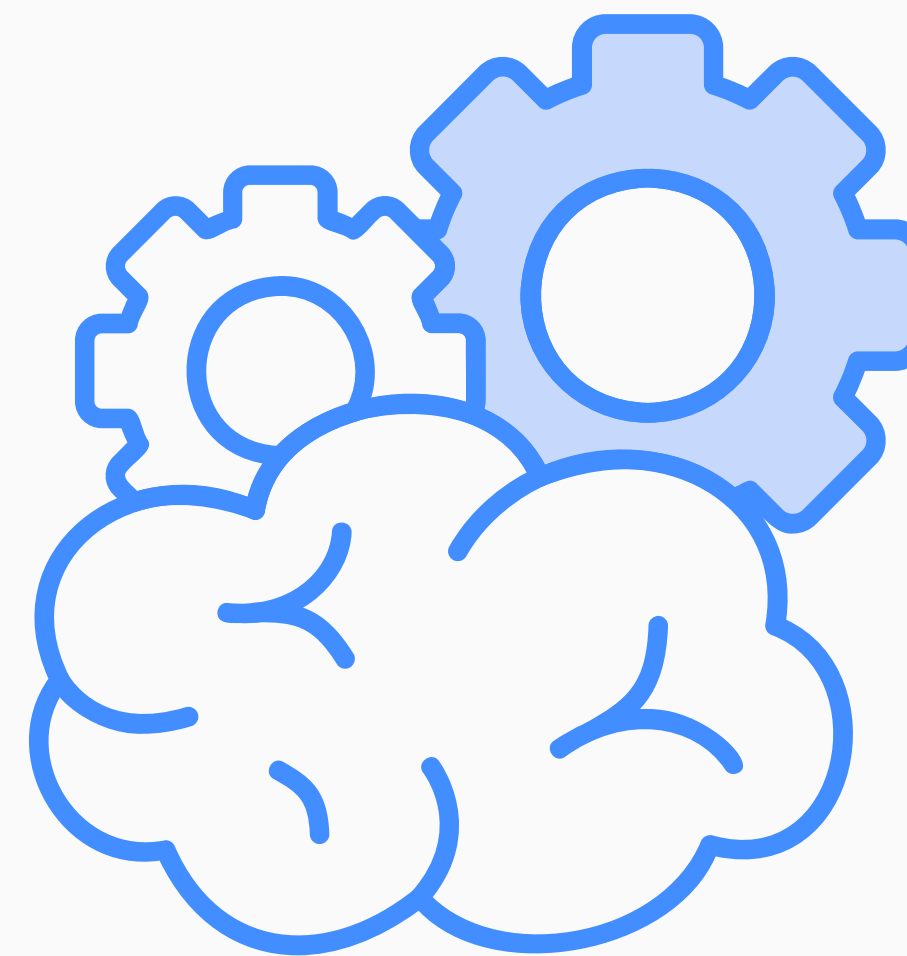
DIABETES PREDICTION

Subject : Machine Learning

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Presented By Group 2

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1. PROBLEM STATEMENT

- The objective of this project is to build machine learning models that predict whether a patient has diabetes based on medical information and compare the performance of Logistic Regression and Random Forest.
- Problem Type: **Classification**

1. PROBLEM STATEMENT

- **Dataset Description:** This Dataset has 14 features and 1000 rows
- **Source:** Kaggle
- **Dataset:** Diabetes Prediction Dataset

Target Variable (CLASS):

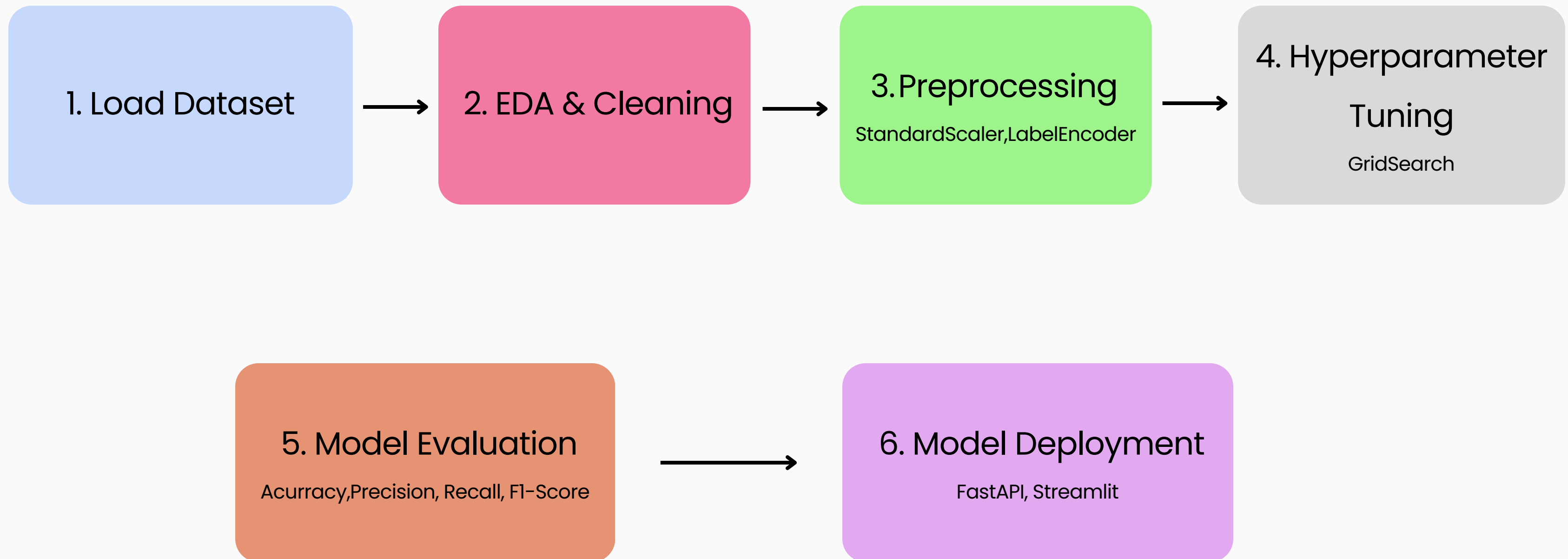
0 = Non-Diabetic

1 = Prediabetic

2 = Diabetic

- **Key Features:** HbA1c, BMI, AGE, Glucose, Urea, Cr (Creatinine), Chol, TG, HDL, LDL, VLDL, Gender.

2. PROJECT WORKFLOW



3. EXPLORATORY DATA ANALYSIS (EDA)

- **Features**

Column	Data type	Description
ID	int64	Unique patient identifier
No_Pation	int64	Patient number/record ID
Gender	object	Male or Female
AGE	int64	Patient's age
Urea	float64	Blood urea level

3. EXPLORATORY DATA ANALYSIS (EDA)

- Features

Column	Data type	Description
Cr	int64	Creatinine level
HbA1c	float64	Average blood glucose level over the past 2–3 months
Chol	float64	Cholesterol level
TG	float64	Triglyceride level
HDL	float64	High-density lipoprotein ("good" cholesterol)

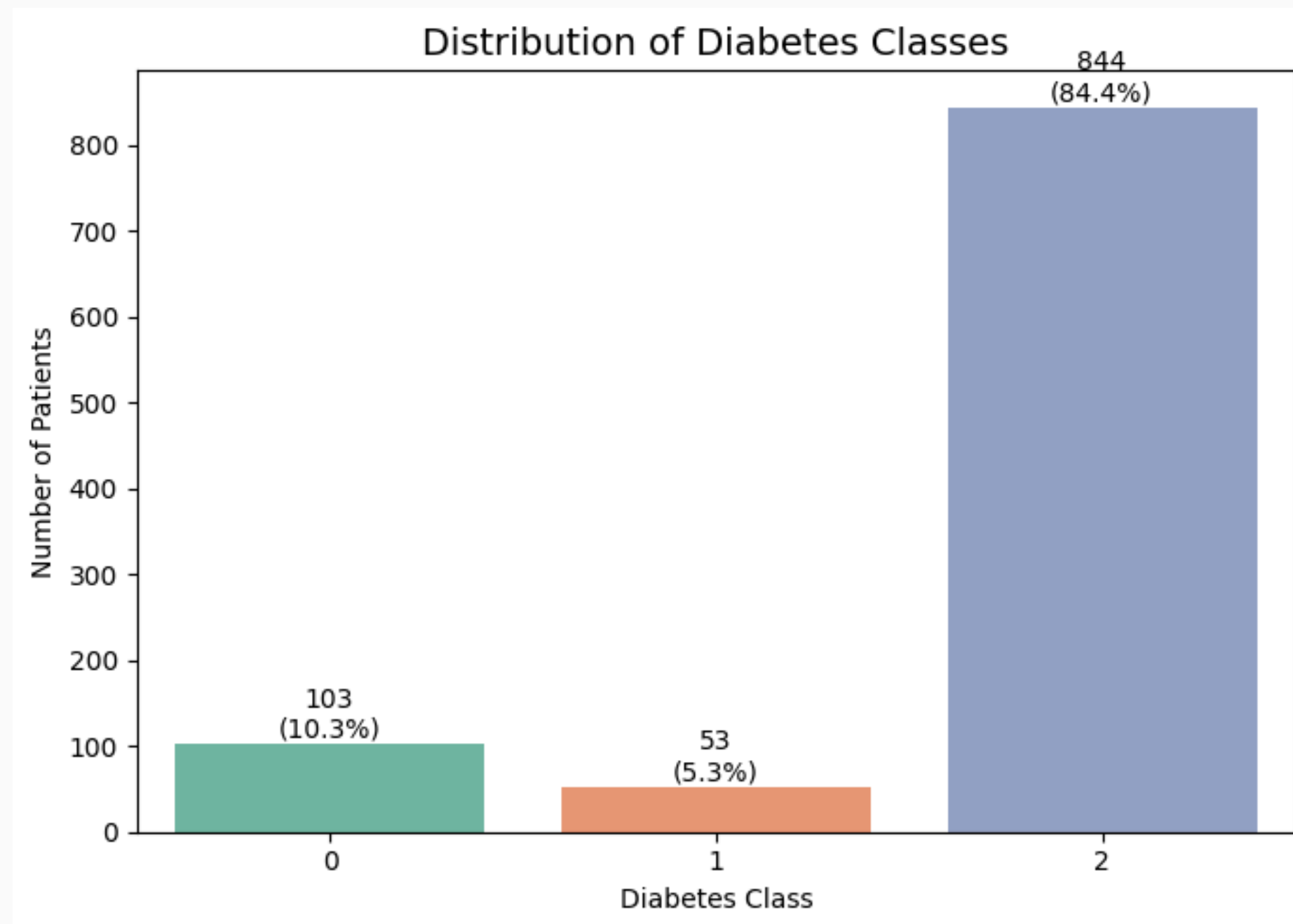
3. EXPLORATORY DATA ANALYSIS (EDA)

- **Features**

Column	Data type	Description
LDL	float64	Low-density lipoprotein ("bad" cholesterol)
VLDL	float64	Very low-density lipoprotein
BMI	float64	Body Mass Index
CLASS	object	Target variable (Diabetic status)

3. EXPLORATORY DATA ANALYSIS (EDA)

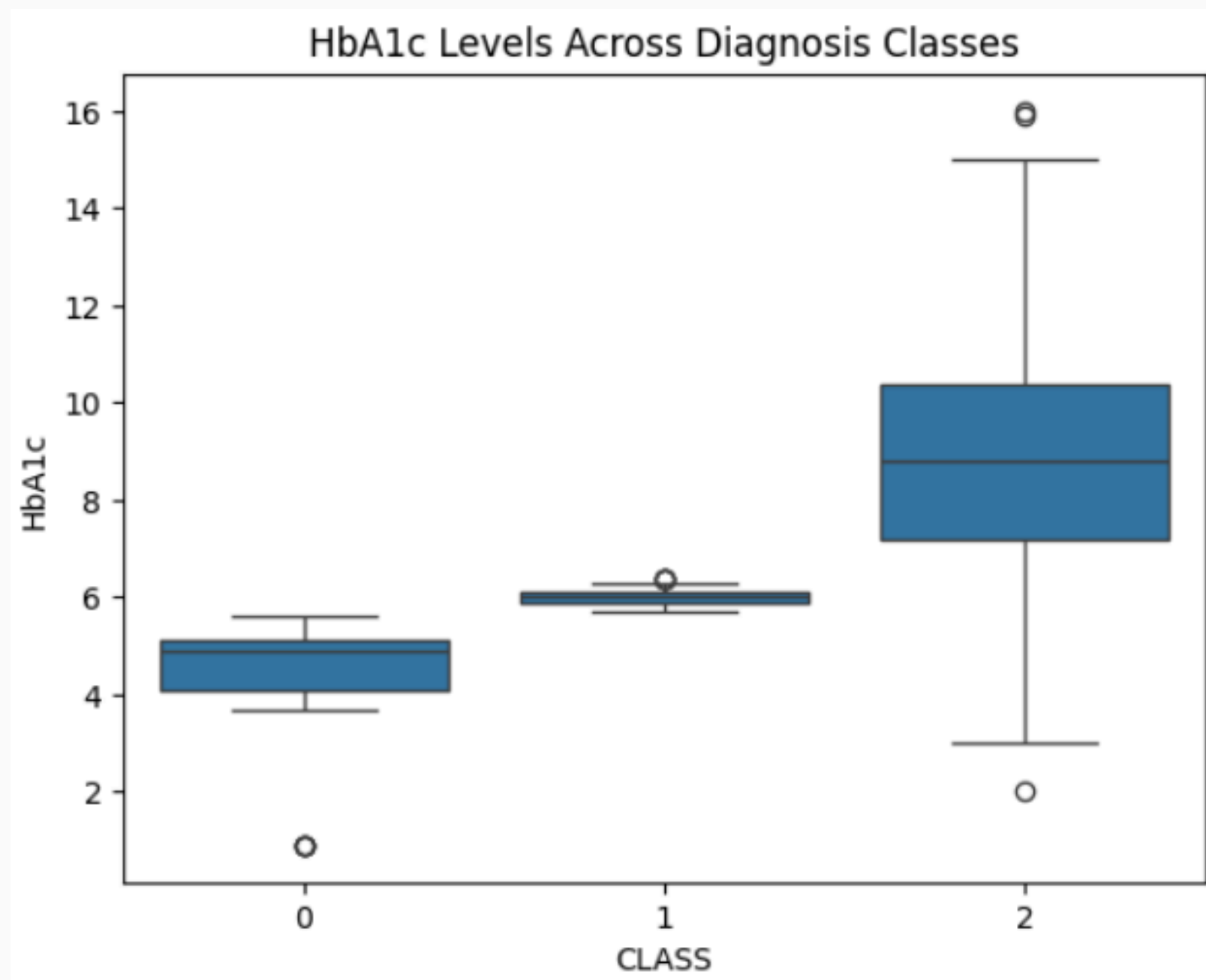
- How many patients are in each diagnosis Class ?



- Class 0 : Non-Diabetic
- Class 1 : Pre-Diabetic
- Class 2 : Diabetic

3. EXPLORATORY DATA ANALYSIS (EDA)

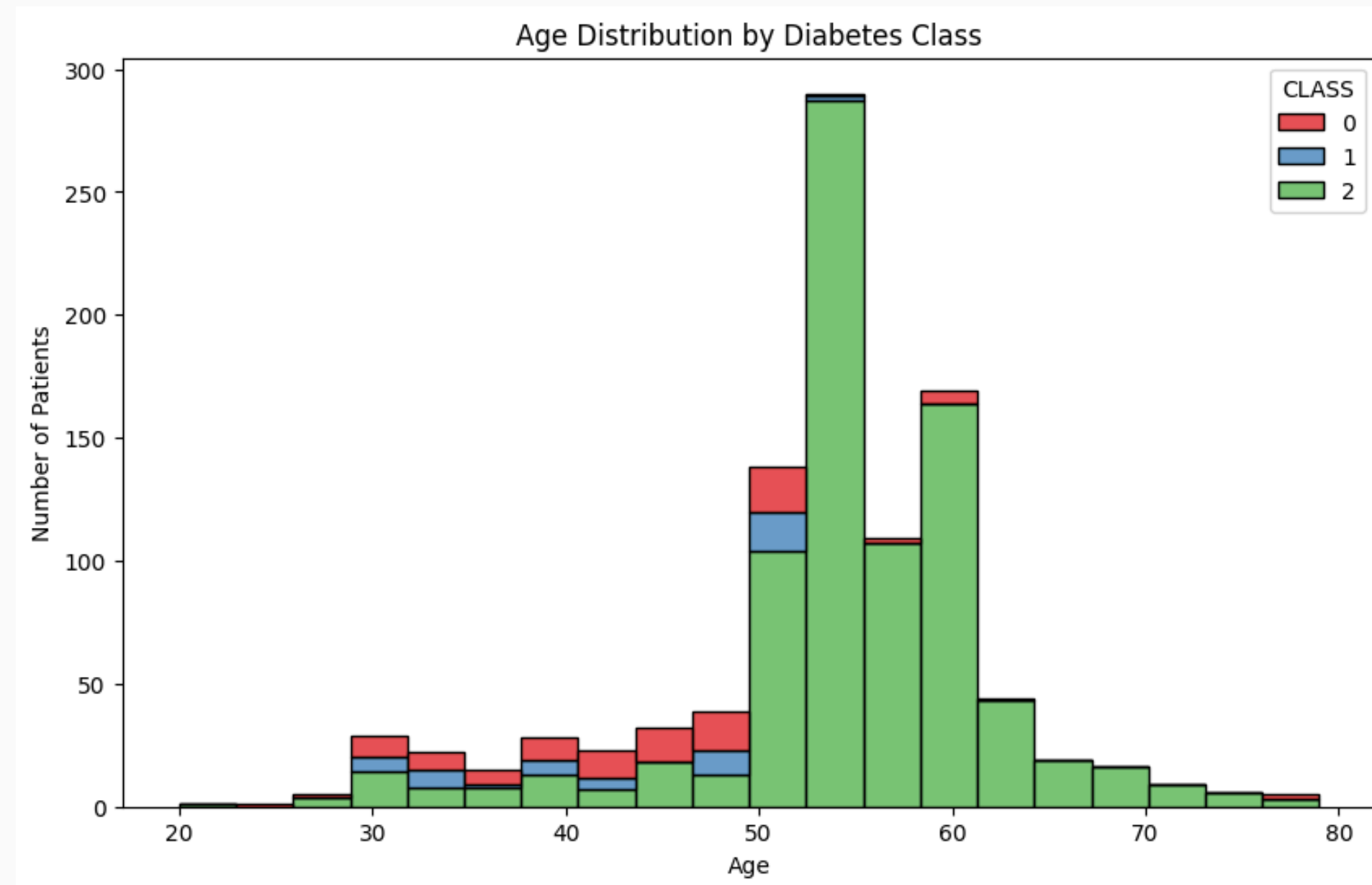
- Which HbA1c levels Range belongs to which Class?



- Class 0 (Non-Diabetic): between 4.0 and 5.2.
- Class 1 (Prediabetic): around 5.7 to 6.3.
- Class 2 (Diabetic): between 7.0 and 10.5 (reaching up to 16).

3. EXPLORATORY DATA ANALYSIS (EDA)

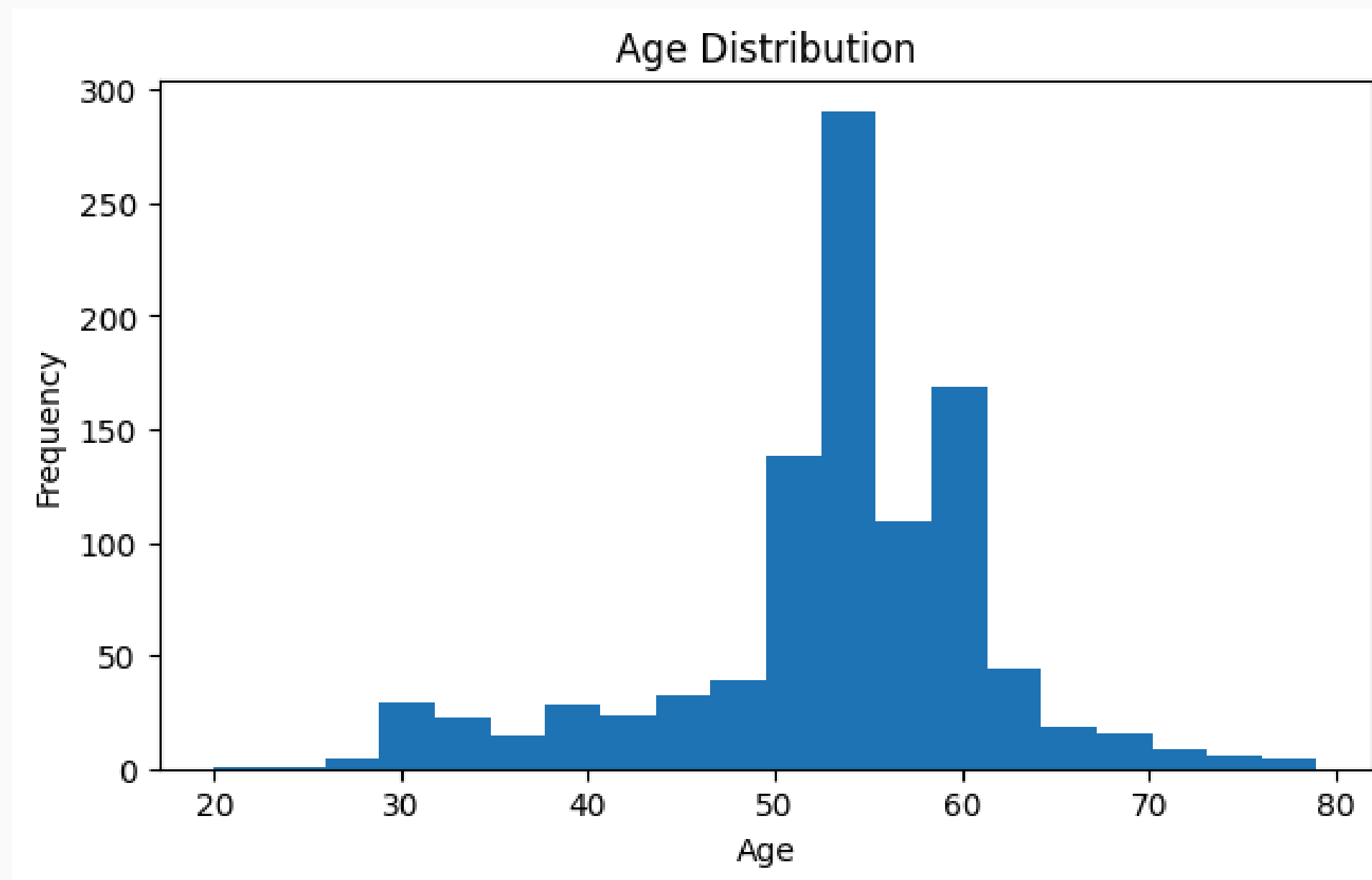
- Which patient's age risk of having diabetes?



- Patients in 50–60 year olds mostly have Diabetes very few patients under 30 or over 70.

3. EXPLORATORY DATA ANALYSIS (EDA)

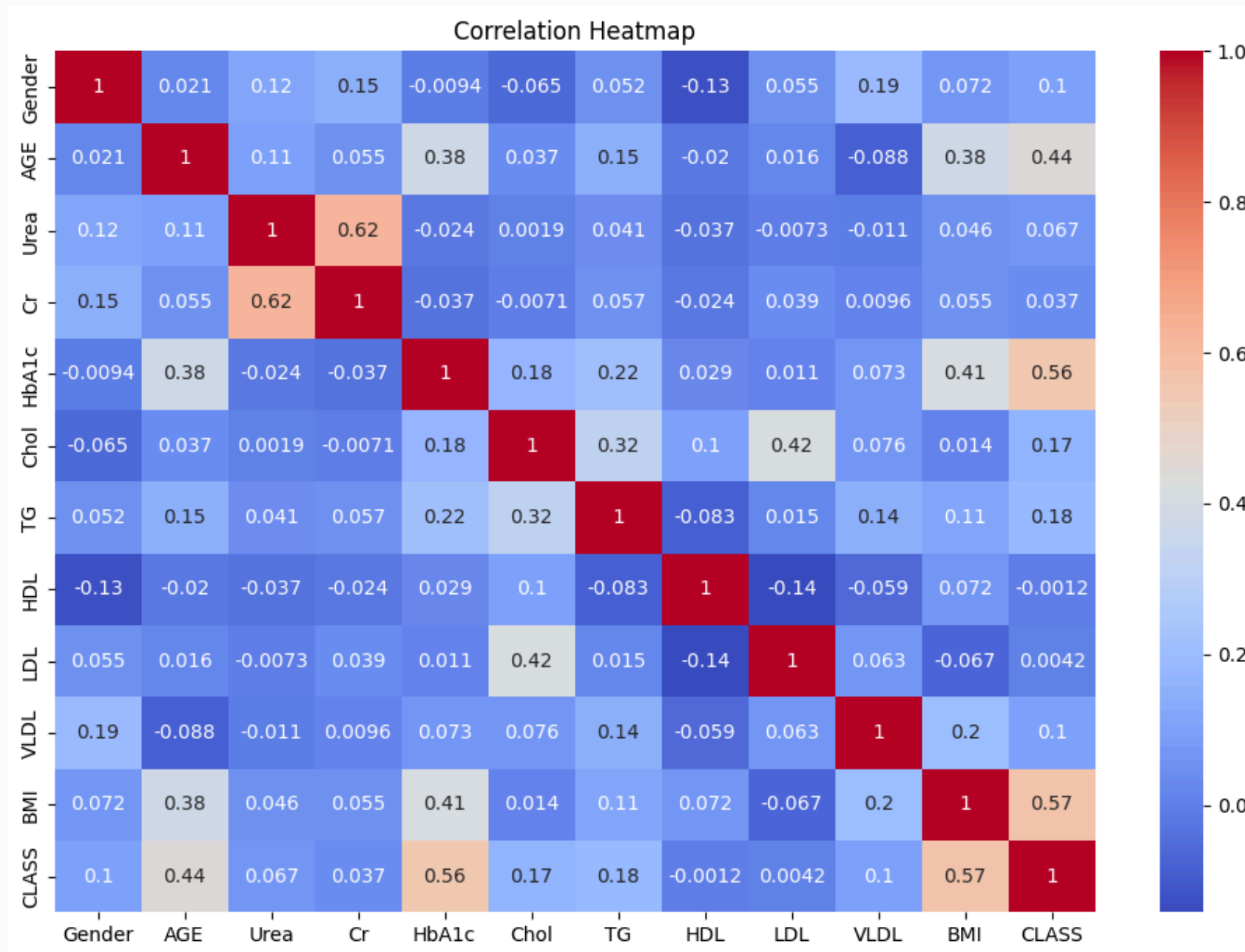
- Which patient's age risk of having diabetes?



- Data is heavily concentrated between ages 50–60 (peaking around 53–55); under 30 and over 70 are rare.

3. EXPLORATORY DATA ANALYSIS (EDA)

- Which features are most strongly tied to diabetes?



1. BMI (Body Mass Index) — Correlation: 0.57

2. HbA1c (%) — Correlation: 0.56

3. AGE — Correlation: 0.44

4. METHODOLOGY

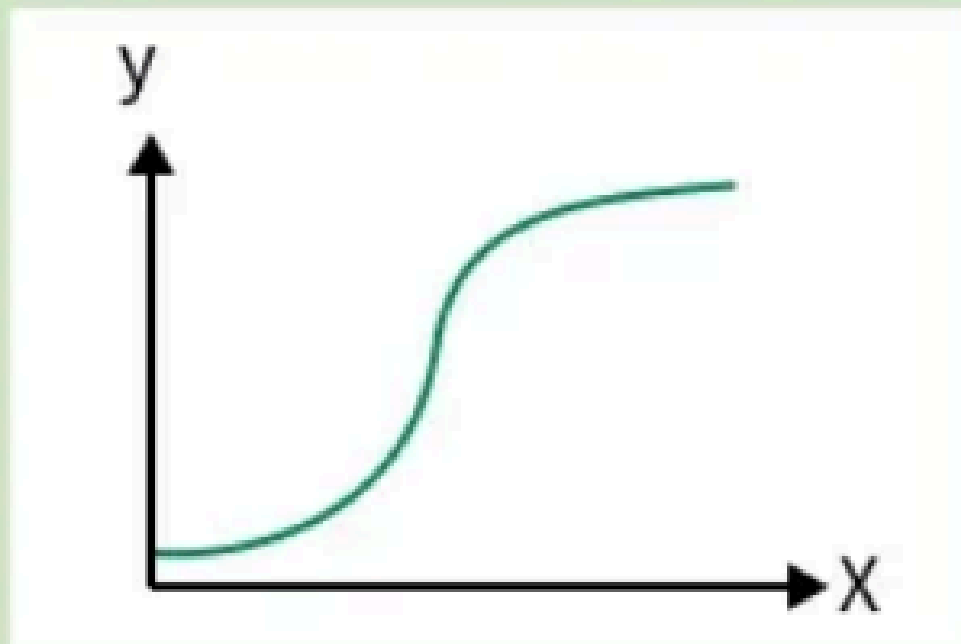
Project Goal: Predict a patient's diabetes status (Non-diabetic, Prediabetic, or Diabetic) using supervised machine learning algorithms to classify the CLASS variable.

1. **Data Cleaning & EDA:** Trimmed whitespace in categorical features and analyzed distributions & boxplots.
2. **Preprocessing:** Encoded categorical variables, dropped non-predictive ID columns, and performed an 80/20 train-test split.
3. **Model Training & Tuning:** Trained Logistic Regression and Random Forest models, optimizing hyperparameters with GridSearchCV.
4. **Evaluation:** Assessed model performance using Accuracy, Precision, Recall, and F1-score.

5. LOGISTIC REGRESSION MODEL

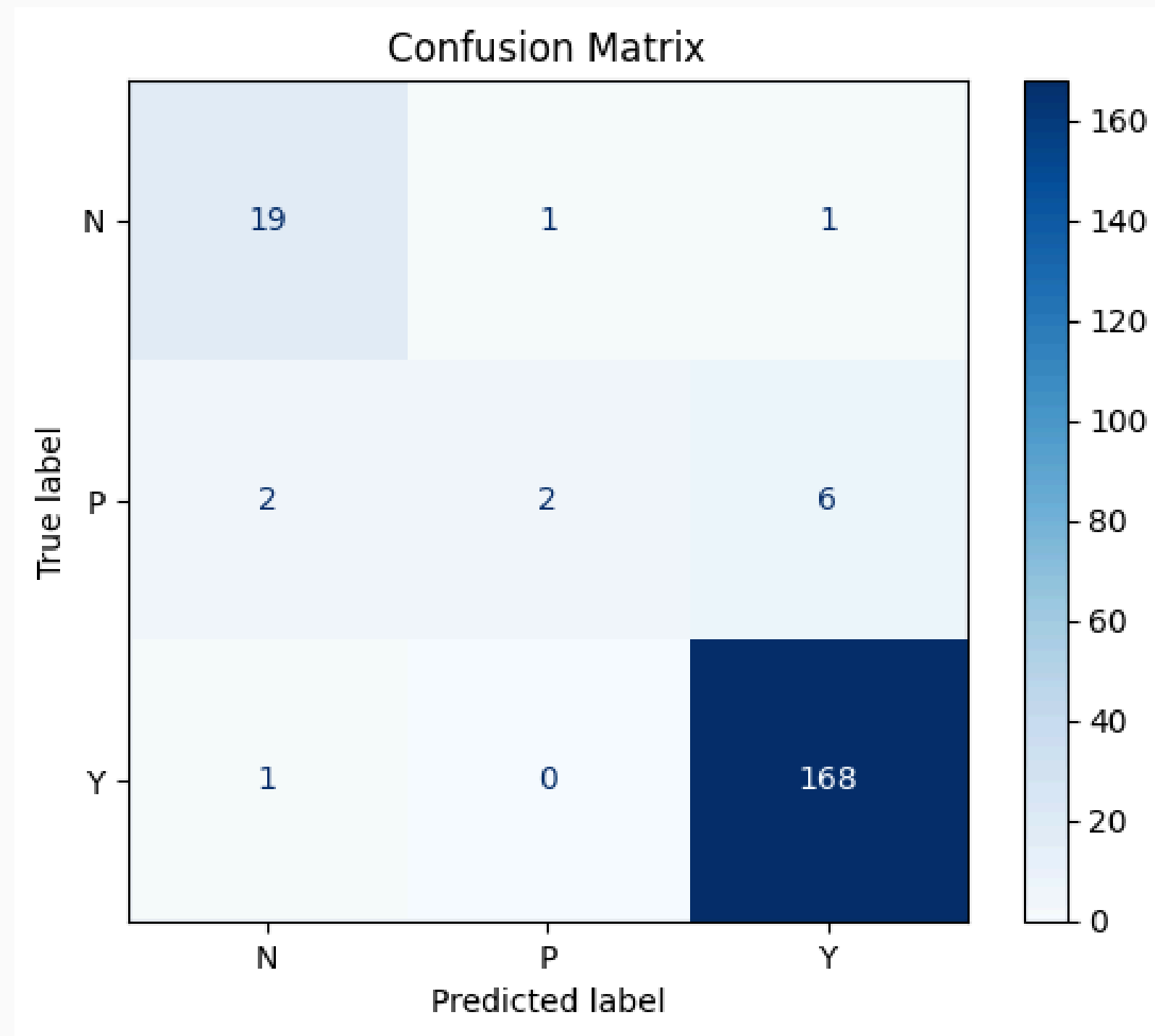
Logistic Regression

- Predicts categorical classes
- Uses sigmoid S-curve
- Solves classification problems



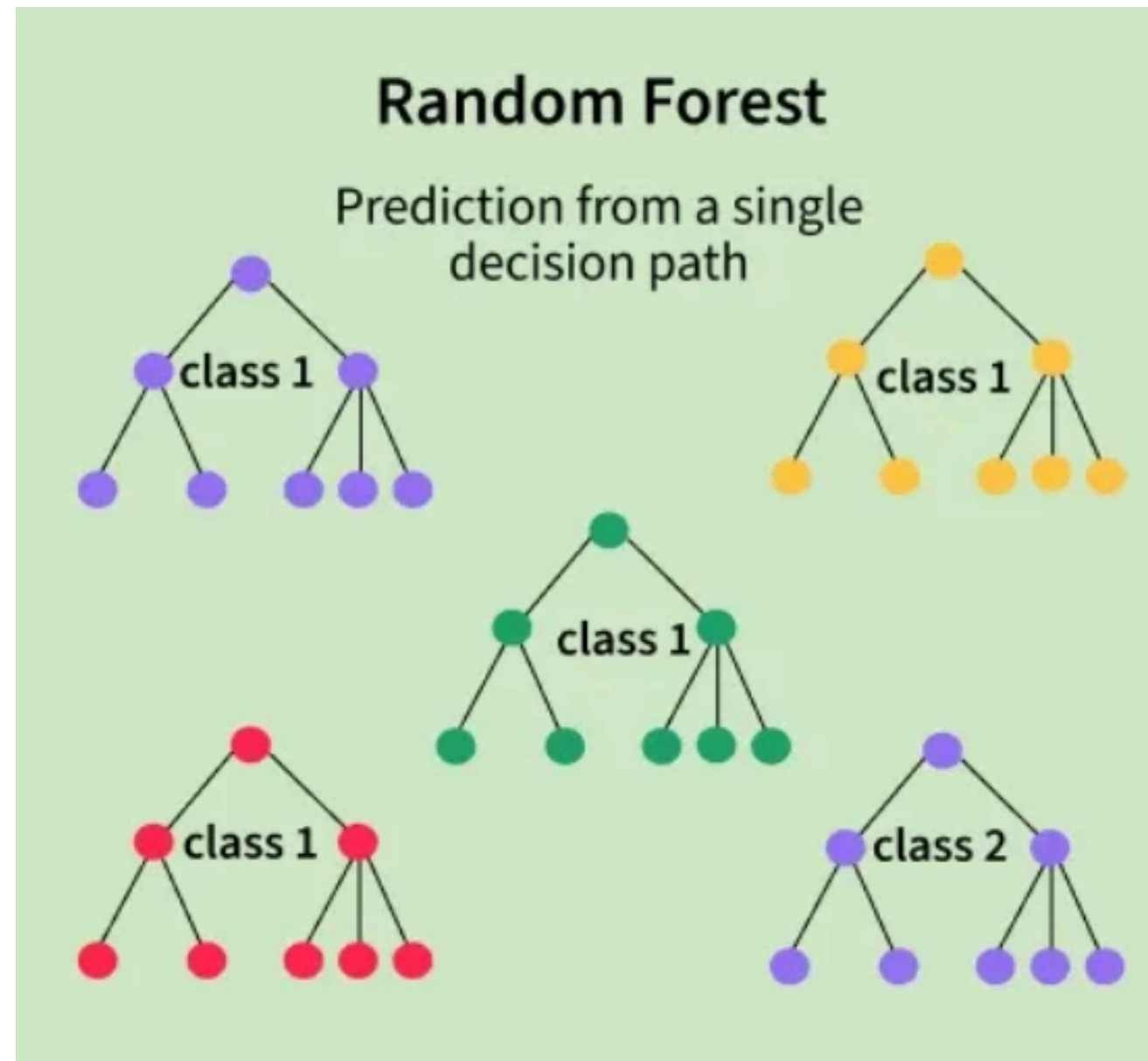
- It predicts the probability that a patient belongs to a specific diabetes class based on the input features and uses the sigmoid (logistic) function to convert the prediction into a probability between 0 and 1. It is simple, fast, and easy to interpret.

5. LOGISTIC REGRESSION MODEL



- The Logistic Regression model correctly classified most patients, especially the Diabetic (Y) class. However, it had difficulty distinguishing Prediabetic (P) patients from Diabetic (Y) patients, leading to several misclassifications.

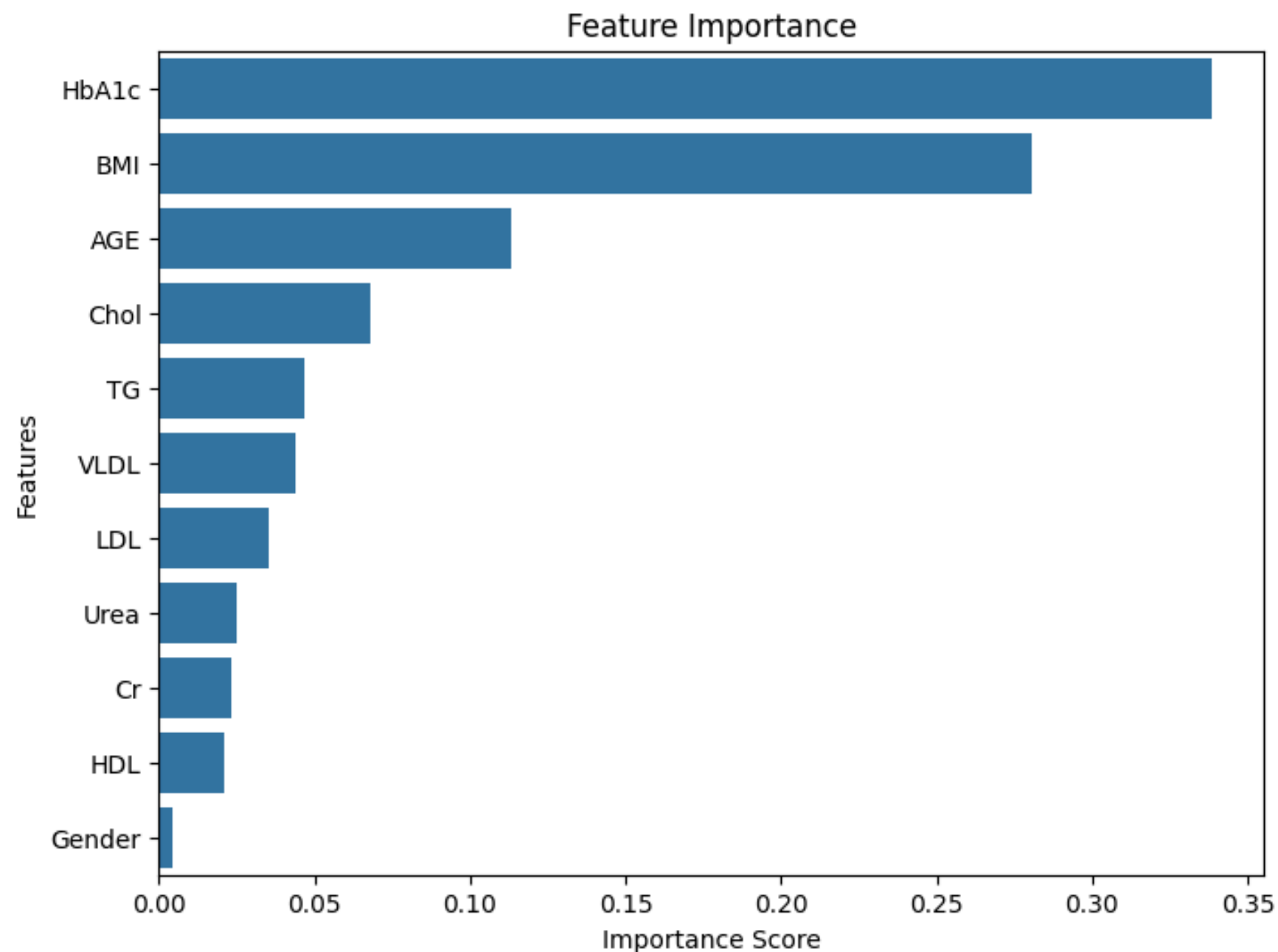
6. RANDOM FOREST MODEL



- Random Forest builds multiple decision trees and combines their predictions using majority voting to make the final classification.
- It is highly accurate, handles complex relationships in the data, is resistant to overfitting, and performs well with medical datasets.

6. RANDOM FOREST MODEL

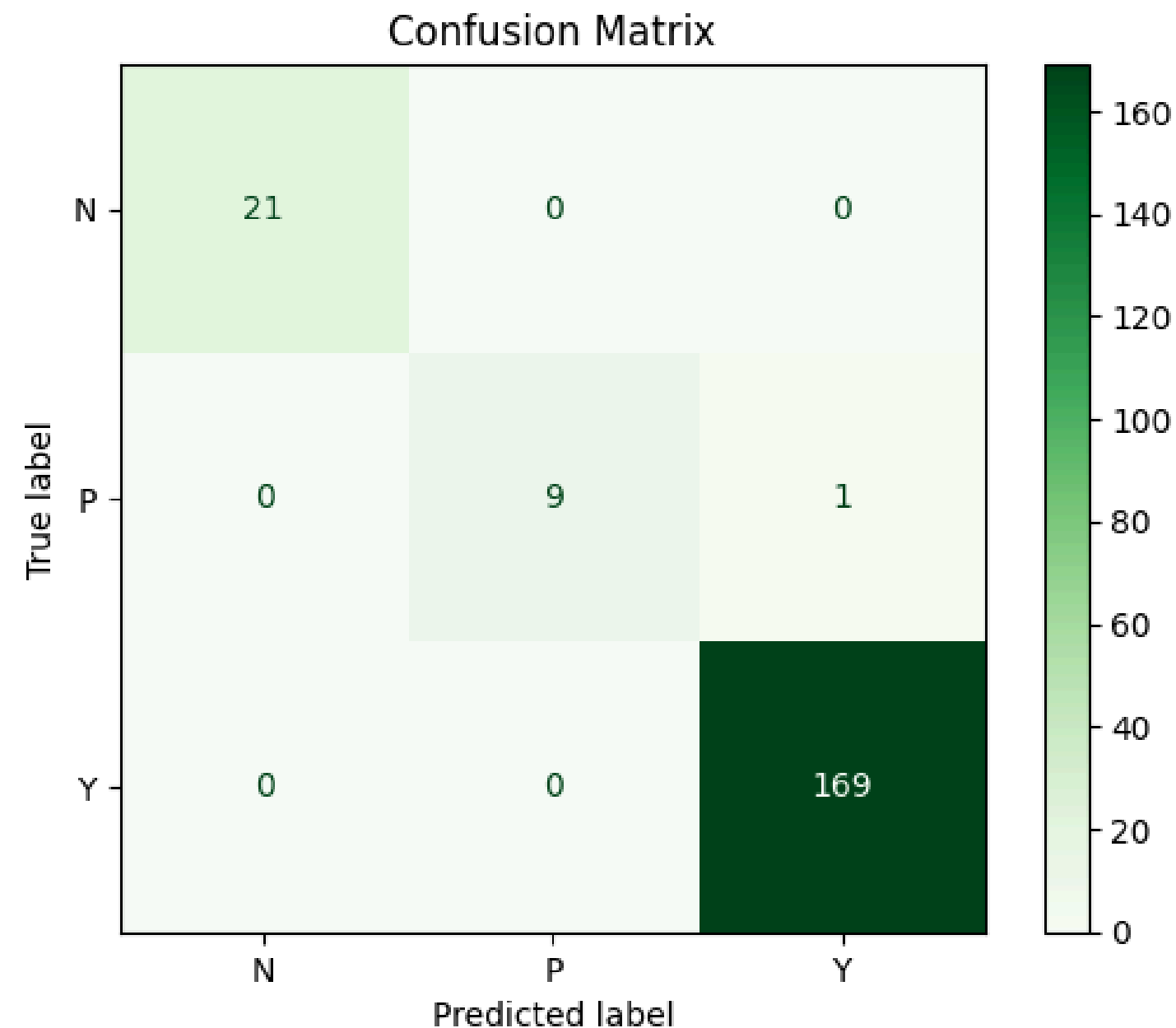
1. Which features are most strongly tied to diabetes?



- HbA1c was the most important feature, followed by BMI and Age. Gender had the least influence. This indicates that blood sugar level (HbA1c), body mass index, and age were the strongest predictors of diabetes in this dataset.

6. RANDOM FOREST MODEL

- Random Forest



- The Random Forest model achieved nearly perfect classification. It correctly predicted every Non-diabetic and Diabetic patient and made only one error by classifying a Prediabetic patient as Diabetic.

7.COMPARATIVE ANALYSIS

- Confusion Matrix comparison

Evaluation	Logistic Regression	Random Forest
Class 0 (Non-Diabetic)	19/21 correct	21/21 correct
Class 1 (Pre-Diabetic)	2/10 correct	9/10 correct
Class 2 (Diabetic)	168/169 correct	169/169 correct
Misclassifications	11	1
Overall Performance	Good	Excellent

7.COMPARATIVE ANALYSIS

Metric	Logistic Regression	Random Forest
Accuracy	94.50%	99.50%
Precision	93.52%	99.50%
Recall	94.50%	99.50%
F1-Score	93.35%	99.49%

8. CHALLENGES & LESSONS LEARNED

- During the project, some challenges were encountered. The dataset contained extra spaces in the **CLASS** and **Gender** columns, which had to be encoded before training the models. It was also important to use different preprocessing steps for each algorithm, such as scaling the data for Logistic Regression but not for Random Forest.
- This project helped improve skills in data preprocessing, exploratory data analysis, model training, hyperparameter tuning, and model evaluation.

9. CONCLUSION

- Both models performed well, but Random Forest achieved the best results, with an accuracy of **99.50%**, compared to **94.50%** for Logistic Regression.
- Based on the results, **Random Forest** is the recommended model for this diabetes prediction task because it provides higher accuracy and better overall performance. The project also demonstrates how machine learning can be used to support early diabetes prediction using patient health data.



**THANK YOU
VERY MUCH**